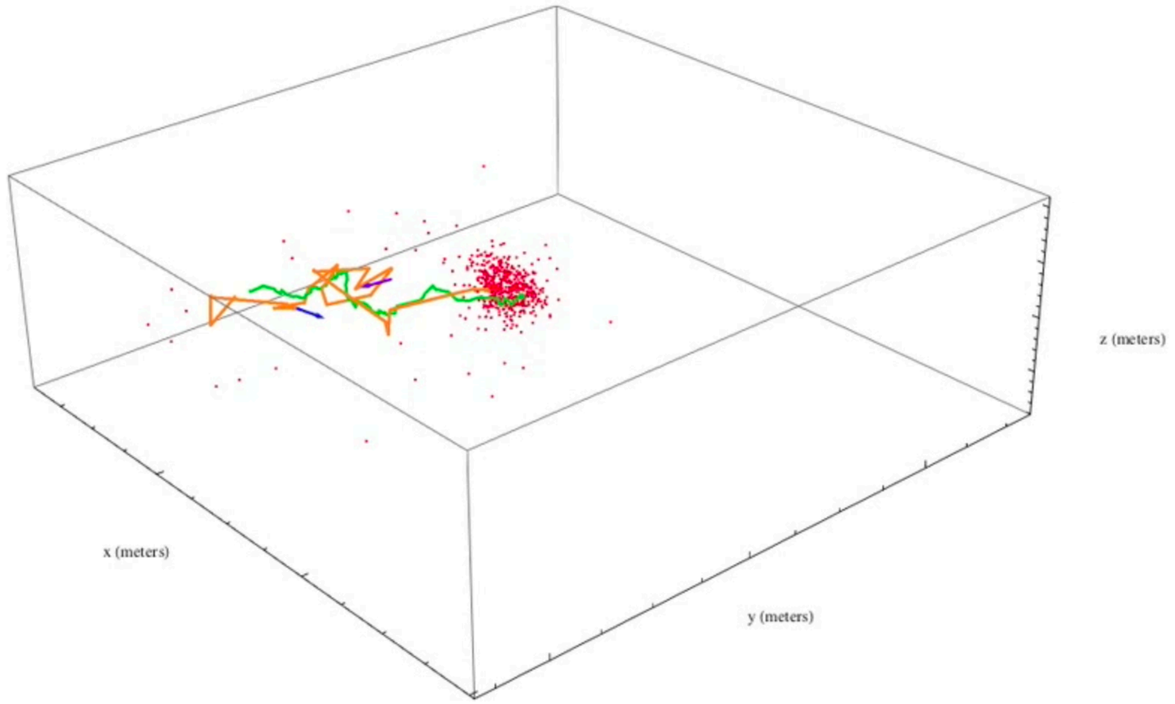
ROB-UY 3303

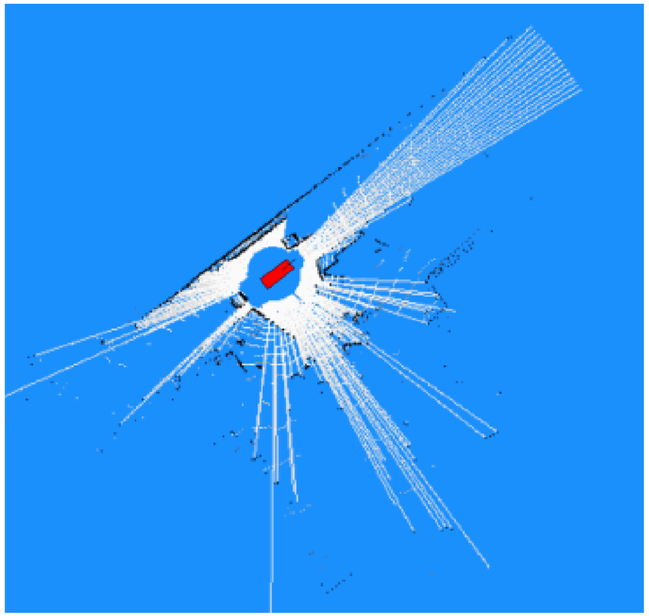
Robot Motion & Planning

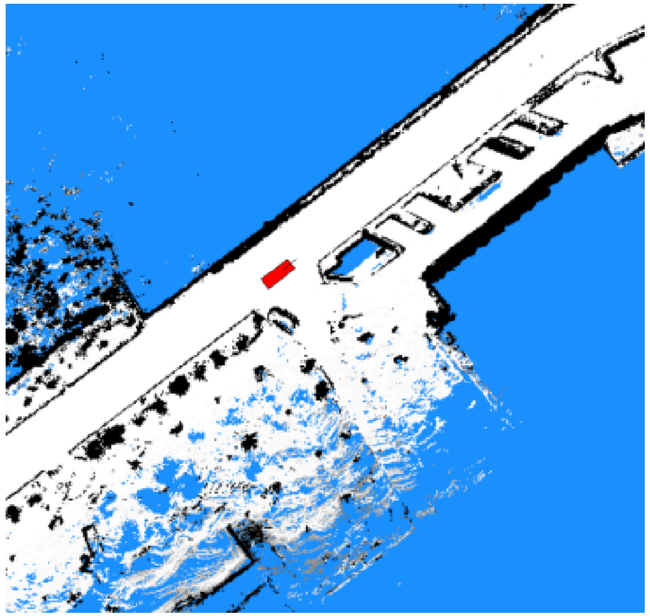
Lecture 04A - PF Localization

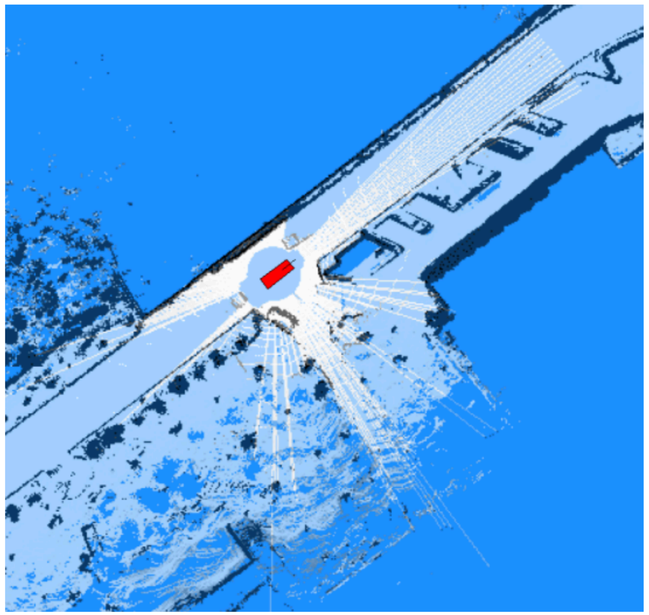


Localization:

Given a map of the world, where are you within that map?







Agenda

1. Particle Representation
2. Particle Filter Algorithm
3. Resampling
4. PF Examples

Agenda

1. **Particle Representation**
2. Particle Filter Algorithm
3. Resampling
4. PF Examples

1. Particles

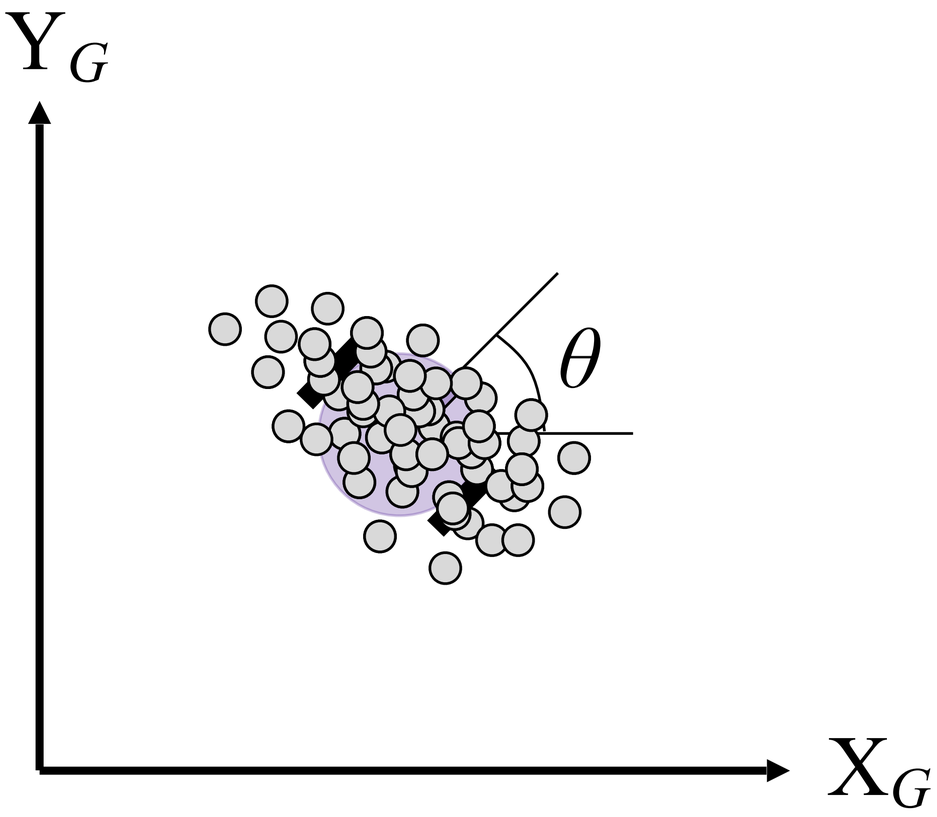
- The belief is represented as a set of particles
- At time t, each particle i, has two key elements:
  - A state vector - X_t^i
  - A weight - w_t^i

$$p_t^i = \begin{bmatrix} X_t^i \\ w_t^i \end{bmatrix}$$

1. Particles

- Each particle is defined by a state and weight. Example:

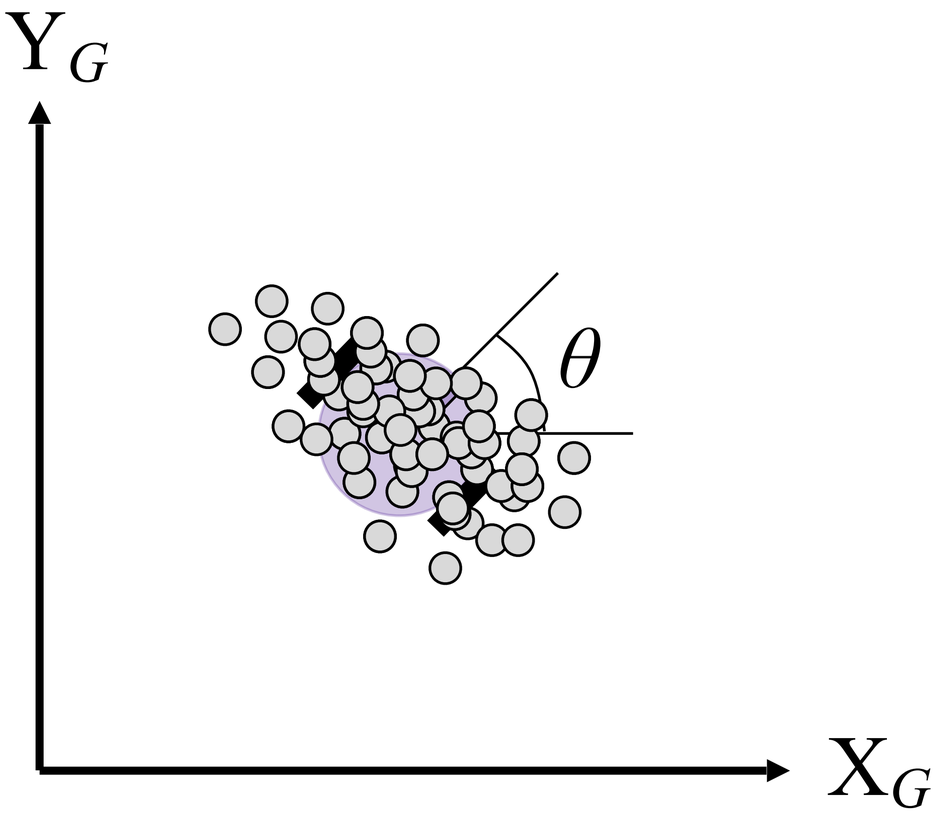
$$p_t^i = \begin{bmatrix} x_t^i \\ y_t^i \\ \theta_t^i \\ w_t^i \end{bmatrix}$$



1. Particles

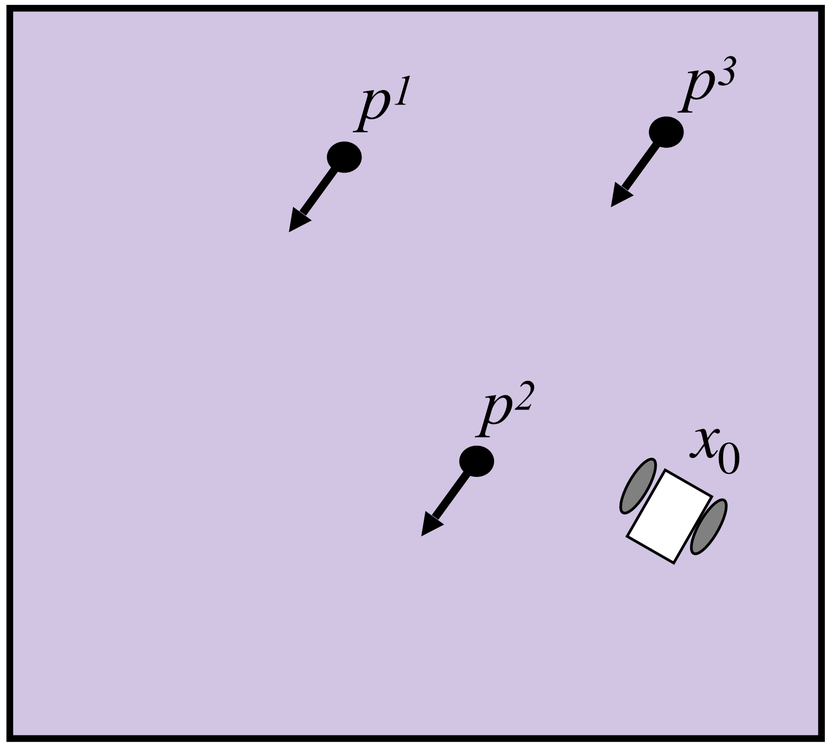
- The belief is represented as a set of n particles.

$$P_t = \begin{bmatrix} p_t^1 \\ p_t^2 \\ \cdots \\ p_t^n \end{bmatrix}$$

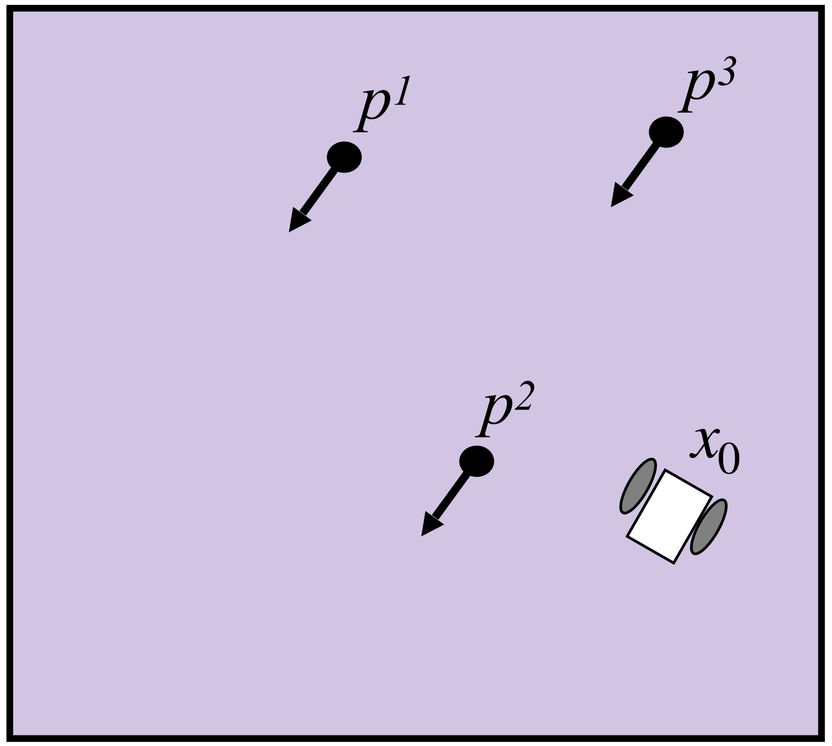


- **Example:**
  - A PF uses *3* particles to represent the position of a (white) robot in a square room.
  - The robot has a perfect compass

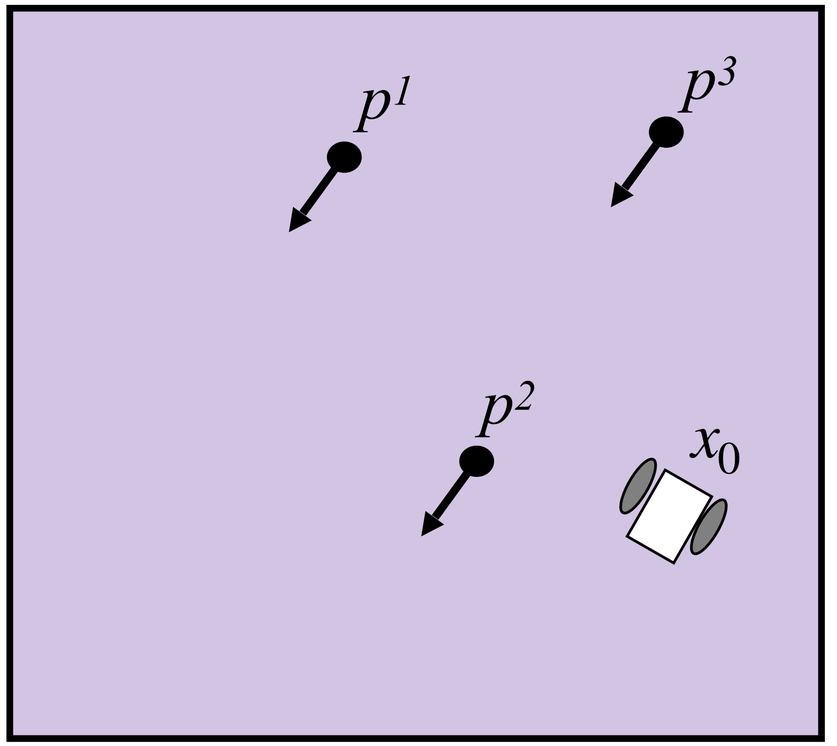
$$p_t^i = \begin{bmatrix} x_t^i \\ y_t^i \\ w_t^i \end{bmatrix}$$



- **Example:**
  - Each of the particles have associated weights.
  - In the example below, p^2 should have the highest weight.



- **How many particles to use?**
  - More particles ensures a higher likelihood of converging to the correct belief state.
  - Fewer particles may be necessary to ensure real-time implementation.



Agenda

1. Particle Representation
2. **Particle Filter Algorithm**
3. Resampling
4. PF Examples

- **Algorithm (Initialize at $t = 0$):**
  - Randomly draw n states in the work space, from a uniform distribution, and add them to the set P_0.

$$P_0 = \begin{bmatrix} p_t^1 \\ p_t^2 \\ \cdots \\ p_t^n \end{bmatrix}$$

  - Iterate on these n states over time (see next slides).

- Particle Filter Algorithm

Algorithm Particle_Filter(P_{t-1}, u_t, z_t)***:***

for $i = 1 \;..\; n$

x_t^i = sample from $P(x_t \mid x_{t-1}, u_t)$

$w_t^i = P(z_t \mid x_t)$

$p_t'^i = [x_t^i, w_t^i]^T$

$P_t' = [p_t'^1, p_t'^2, \ldots, p_t'^n]$

for $i = 1 \;..\; n$

p_t^i = sample from P_t' given weights

$P_t = [p_t^1, p_t^2, \ldots, p_t^n]$

return P_t

- Particle Filter Algorithm

Algorithm Particle_Filter(P_{t-1}, u_t, z_t)***:***

for $i = 1 \;..\; n$

x_t^i = sample from $P(x_t \mid x_{t-1}, u_t)$

$w_t^i = P(z_t \mid x_t)$

$p_t'^i = [x_t^i, w_t^i]^T$ Prediction Step

$P_t' = [p_t'^1, p_t'^2, \ldots, p_t'^n]$

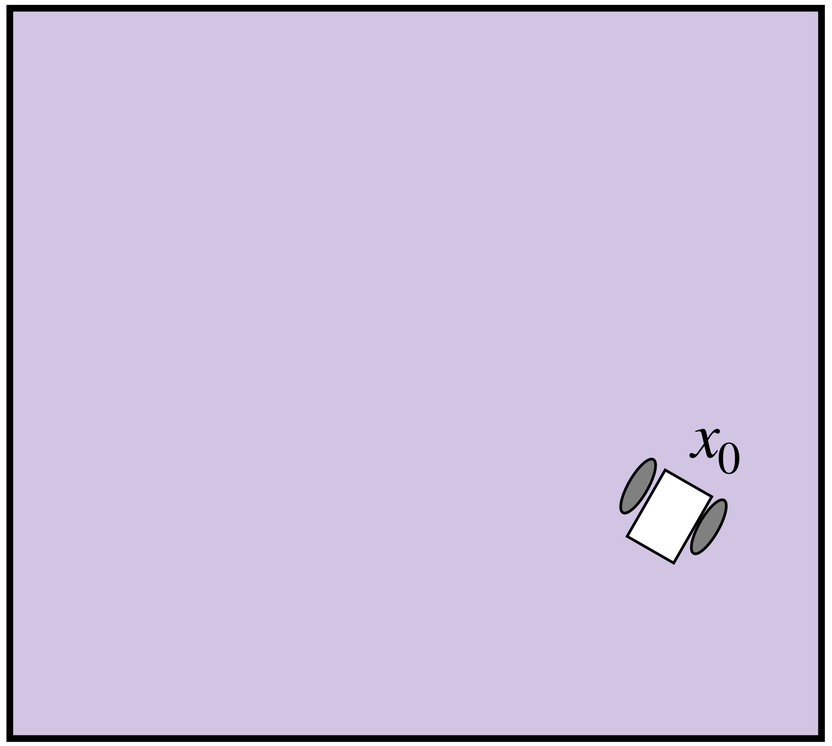
for $i = 1 \;..\; n$ Correction Step

p_t^i = sample from P_t' given weights

$P_t = [p_t^1, p_t^2, \ldots, p_t^n]$

return P_t

- **Algorithm Example**
  - Provided is an example where a robot starts at some unknown location in the bounded workspace.

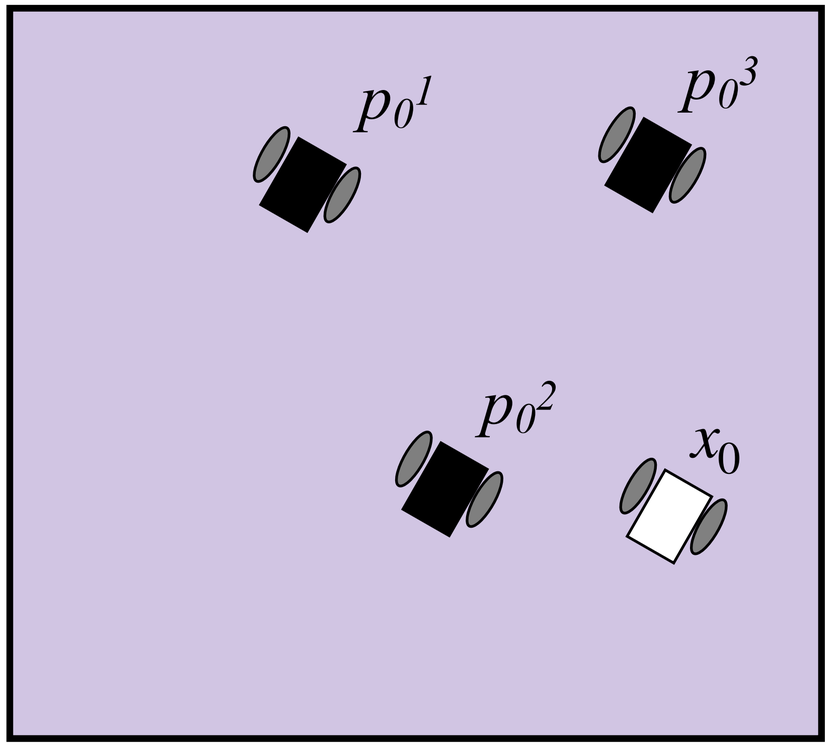


Algorithm Example

- At time step t_0:
  - We randomly pick $n = 3$ states represented as:

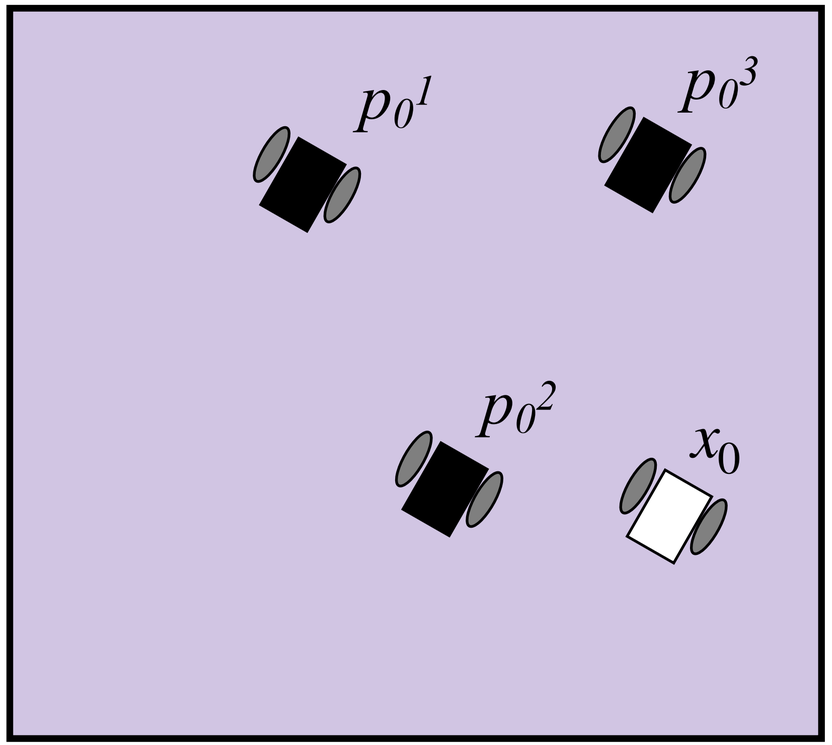
$$P_0 = [p_0^1, p_0^2, p_0^3]^T$$

  - For simplicity, assume known heading



Algorithm Example

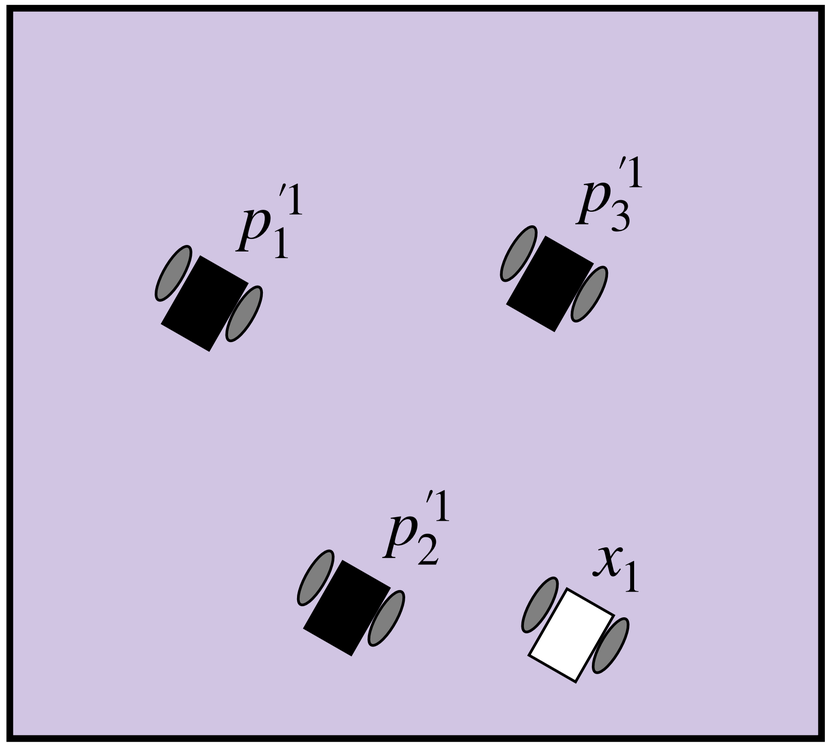
- The next few slides provide an example of one iteration of the algorithm, given P_0.
  - This iteration is time step t_1.
  - The inputs are measurement z_1, odometry $u_1 = [\Delta s_r \ \Delta s_l]$



Algorithm Example

- **Time step t_1**
  - Prediction: Randomly generate new states by propagating previous states X_0 with u_1 and random noise.

$$P'_1 = \begin{bmatrix} p_1^{'1} \\ p_2^{'1} \\ p_3^{'1} \end{bmatrix}$$

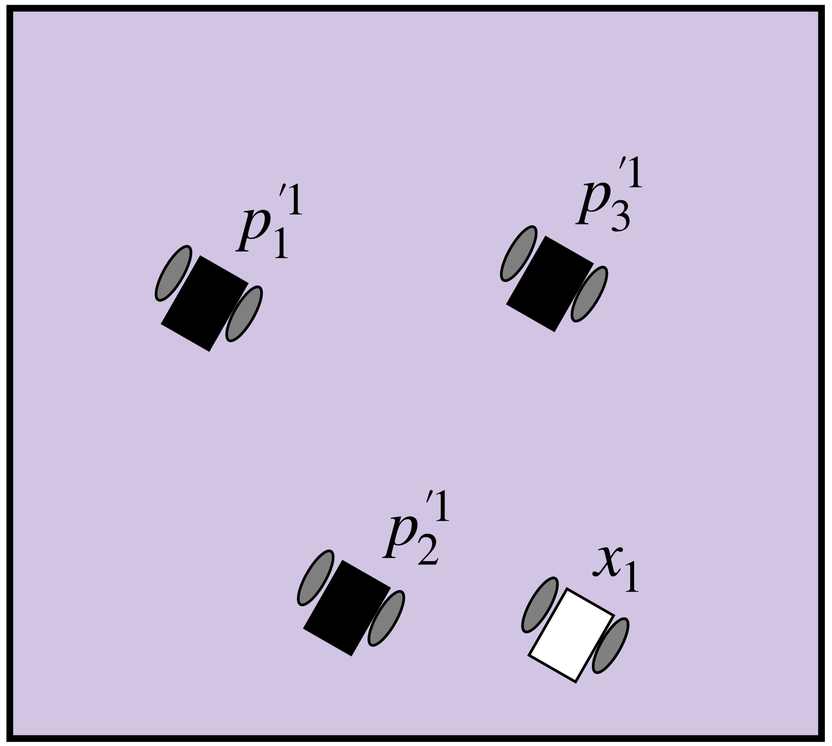


Algorithm Example

- **Time step t_1**
  - Prediction: We use a motion model to generate these predicted states.

$$\Delta x = \Delta s \ \ cos(\theta + \Delta\theta/2)$$

$$\Delta y = \Delta s \ \ sin(\theta + \Delta\theta/2)$$



Algorithm Example

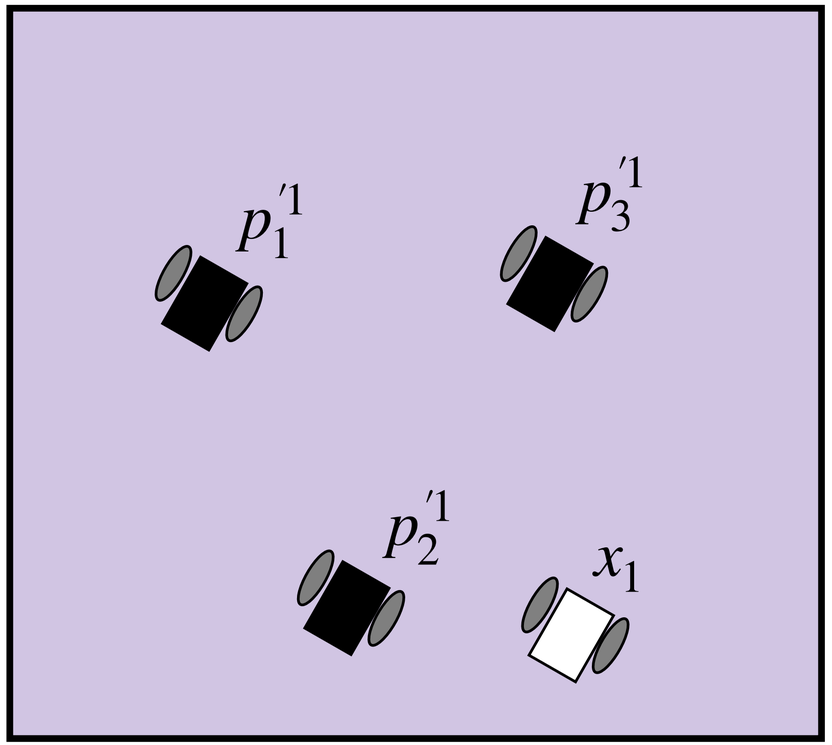
- **Time step t_1**
  - Prediction: We use a motion model to generate these predicted states.

$$\Delta x = \Delta s \ \ cos(\theta + \Delta\theta/2)$$

$$\Delta y = \Delta s \ \ sin(\theta + \Delta\theta/2)$$

$$\Delta s = \frac{\Delta s_r + \Delta s_l}{2}$$

$$\Delta\theta = \frac{\Delta s_r - \Delta s_l}{b}$$



Algorithm Example

- **Time step t_1**
  - Prediction: We use a motion model to generate these predicted states.

$$\Delta x'^i = \Delta s'^i \, cos(\theta'^i + \Delta\theta'^i/2)$$

$$\Delta y'^i = \Delta s'^i \, sin(\theta'^i + \Delta\theta'^i/2)$$

$$\Delta s'^i = \frac{\Delta s_r'^i + \Delta s_l'^i}{2}$$

$$\Delta\theta'^i = \frac{\Delta s_r'^i - \Delta s_l'^i}{b}$$

$$\Delta s_r'^i \sim N(\Delta s_r, \sigma_{sr})$$

$$\Delta s_l'^i \sim N(\Delta s_l, \sigma_{sl})$$

Algorithm Example

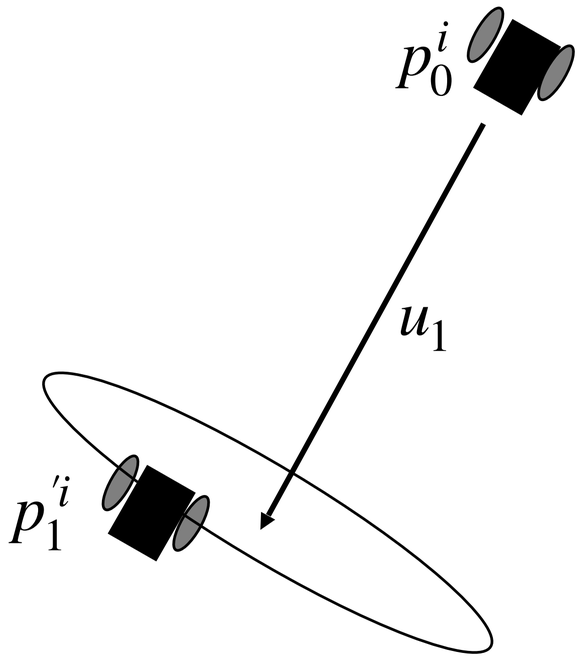
- **Time step t_1**
  - Prediction: We use a motion model to generate these predicted states.

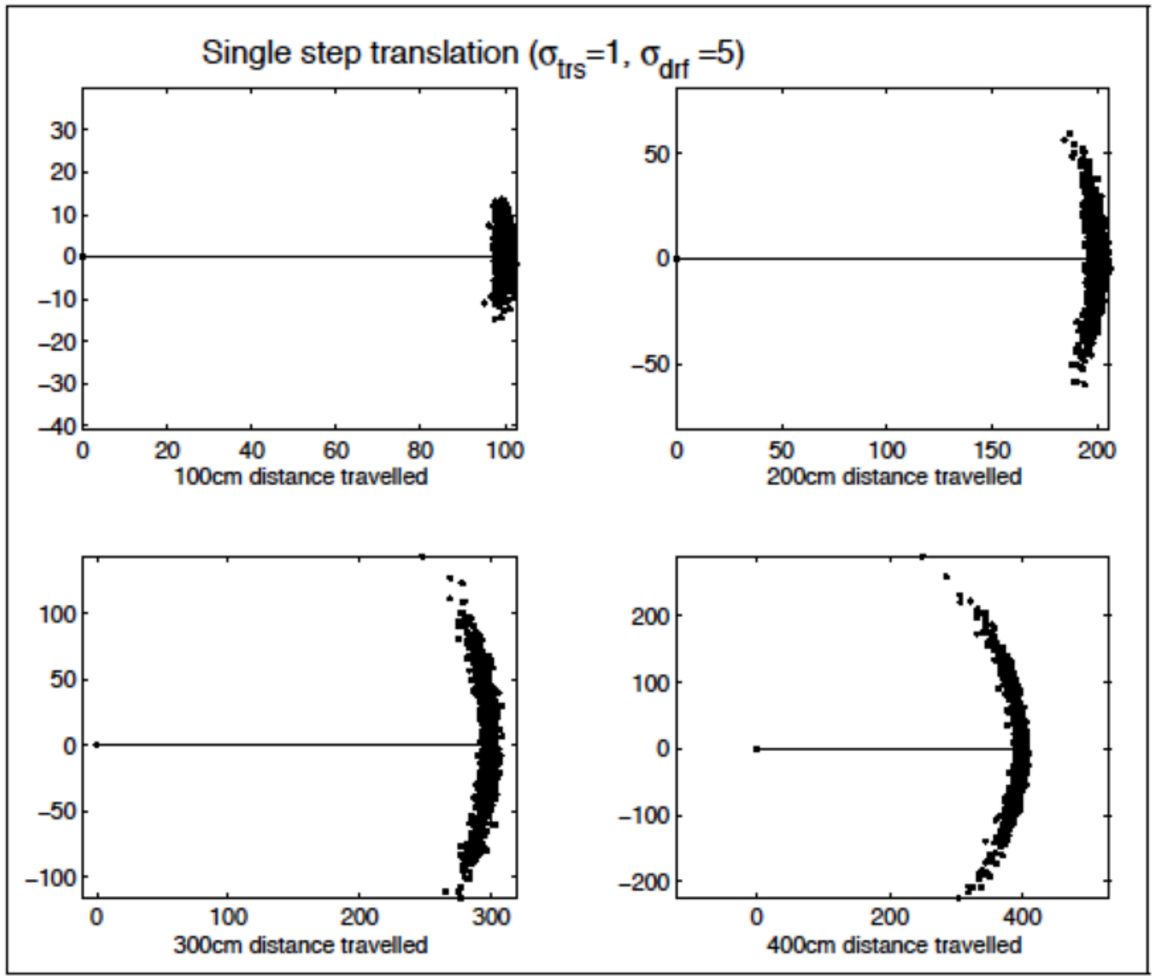
$$\Delta x'^i = \Delta s'^i \; cos(\theta'^i + \Delta\theta'^i/2)$$

$$\Delta y'^i = \Delta s'^i \; sin(\theta'^i + \Delta\theta'^i/2)$$

$$\Delta s'^i = \frac{\Delta s_r'^i + \Delta s_l'^i}{2}$$

$$\Delta\theta'^i = \frac{\Delta s_r'^i - \Delta s_l'^i}{b}$$

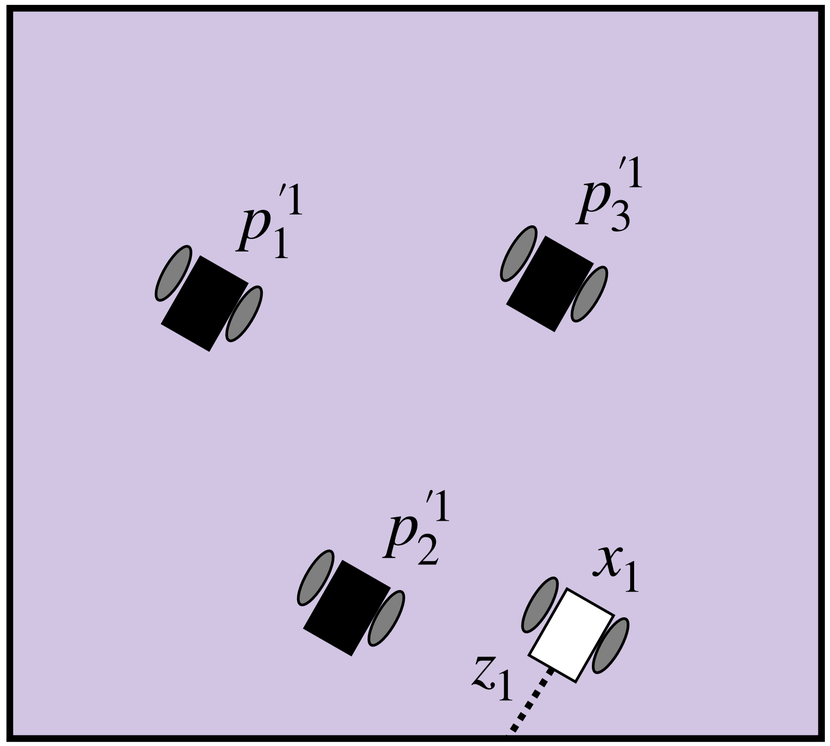




Yiannis, McGill University, PF Tutorial

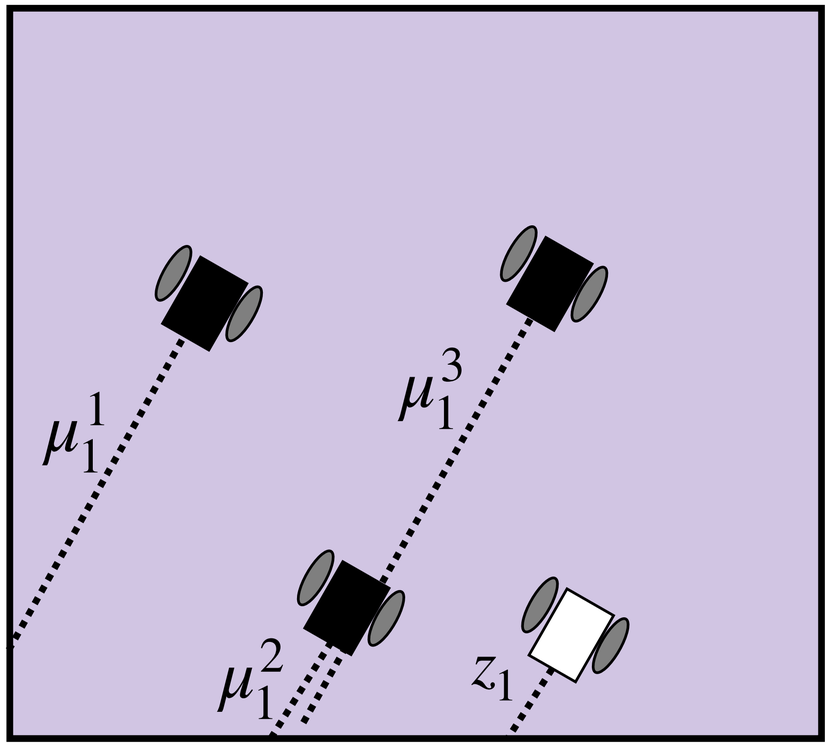
Algorithm Example

- **Time step t_1**
  - Correction: We get a new measurement z_1, e.g. a forward facing range measurement.



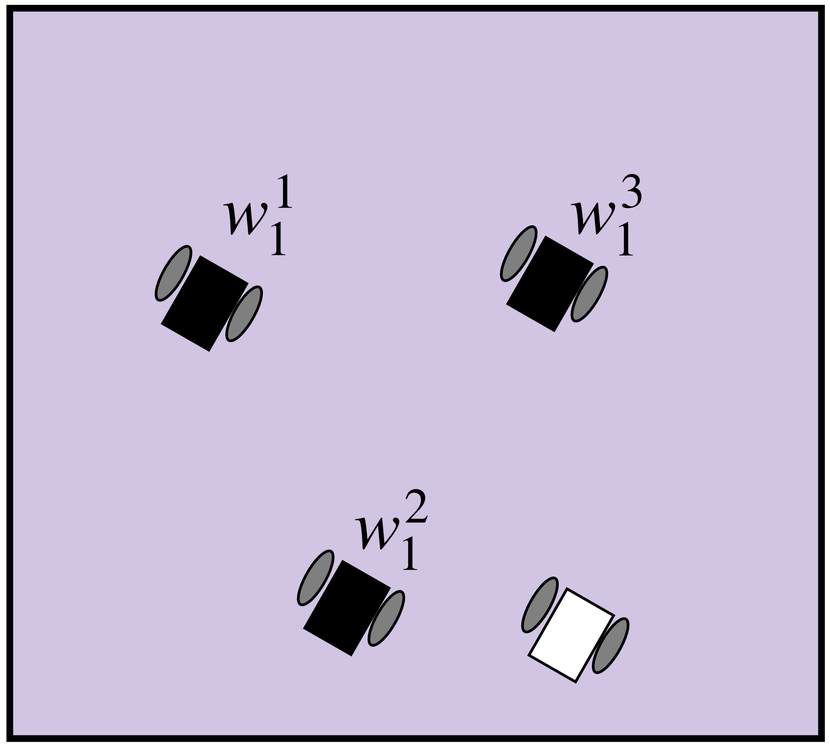
Algorithm Example

- **Time step t_1**
  - Correction: Using the measurement z_1, and expected measurements μ_1^i, calculate the weights $w^i = P(z_1 \mid x_1^{'i})$ for each particle.



Algorithm Example

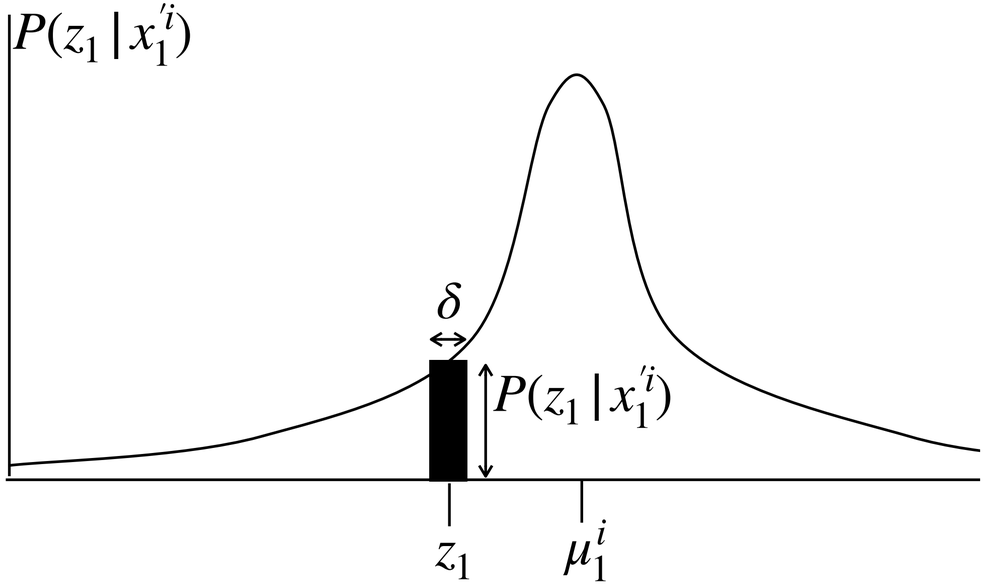
- **Time step t_1**
  - Correction: Using the measurement z_1, and expected measurements μ_1^i, calculate the weights $w^i = P(z_1 \mid x_1^{'i})$ for each particle.



Algorithm Example

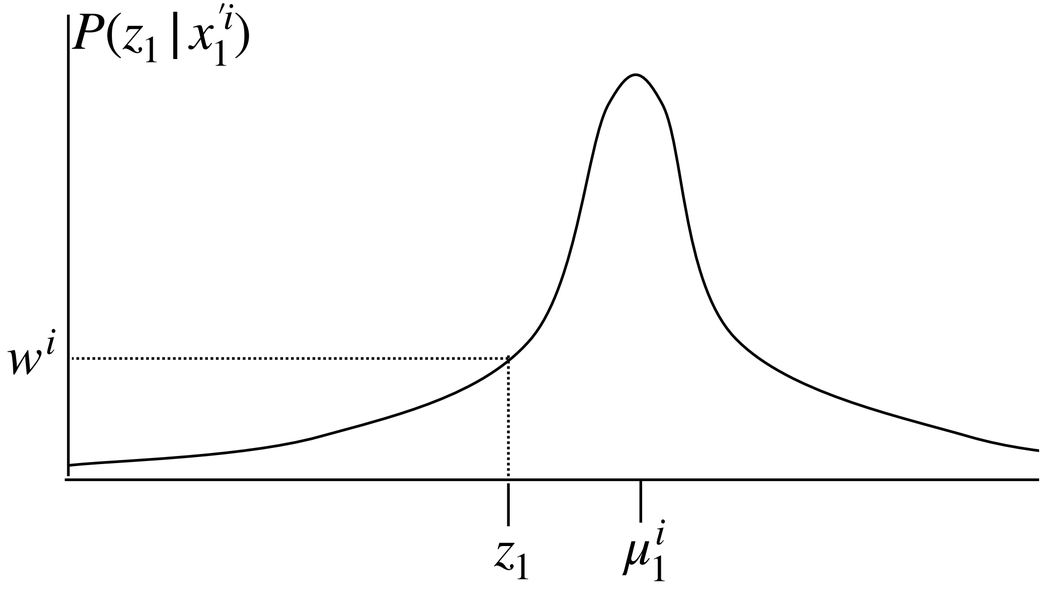
- **Time step t_1**
  - Correction: Using the measurement z_1, and expected measurements μ_1^i, calculate the weights $w^i = P(z_1 \mid x_1^{'i})$ for each particle.

$$w^i = P(z_1 \mid x_1^{'i})\delta$$



Algorithm Example

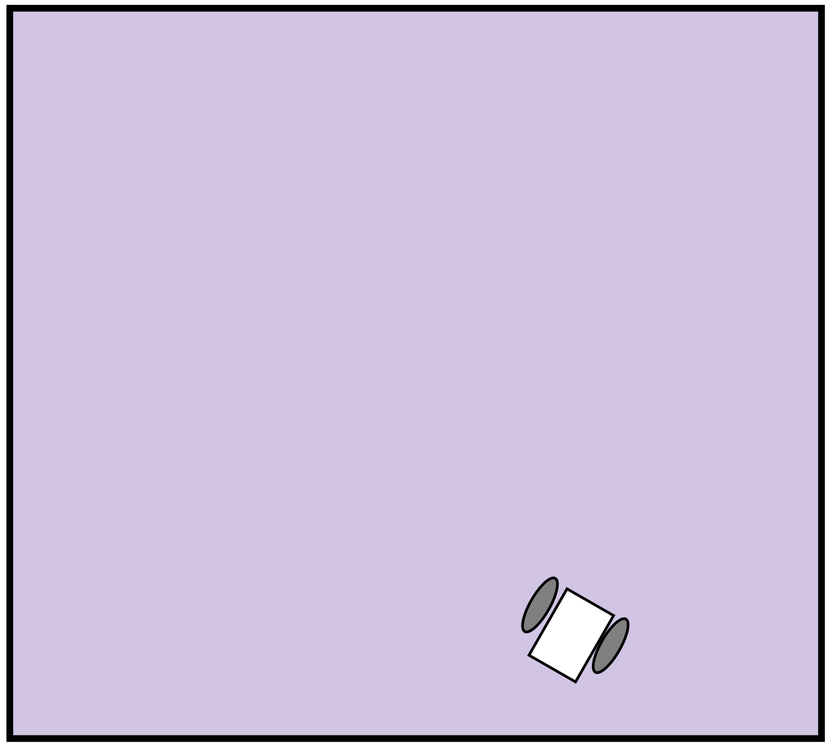
- **Time step t_1**
  - Correction: Using the measurement z_1, and expected measurements μ_1^i, calculate the weights $w^i = P(z_1 \mid x_1^{'i})$ for each particle.
  - In practice, all weights will have the multiplier δ, and they will divide out. So we can ignore them



Algorithm Example

- **Time step t_1**
  - Correction: Resample from the temporary state distribution based on the weights $w_1^2 > w_1^1 > w_1^3$

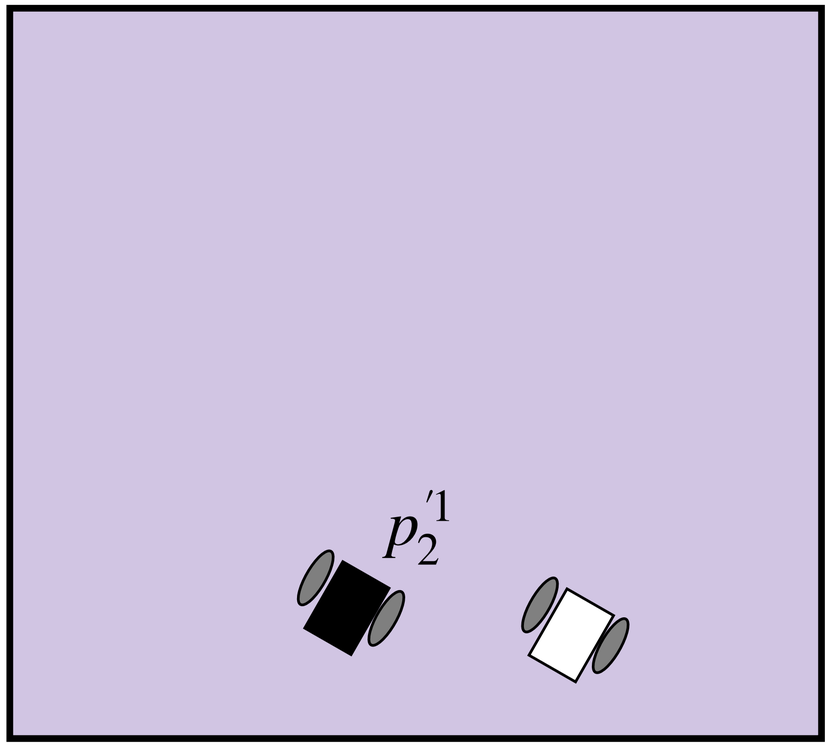
$$P_1 = \begin{bmatrix} & \\ & \end{bmatrix}$$



Algorithm Example

- **Time step t_1**
  - Correction: Resample from the temporary state distribution based on the weights $w_1^2 > w_1^1 > w_1^3$

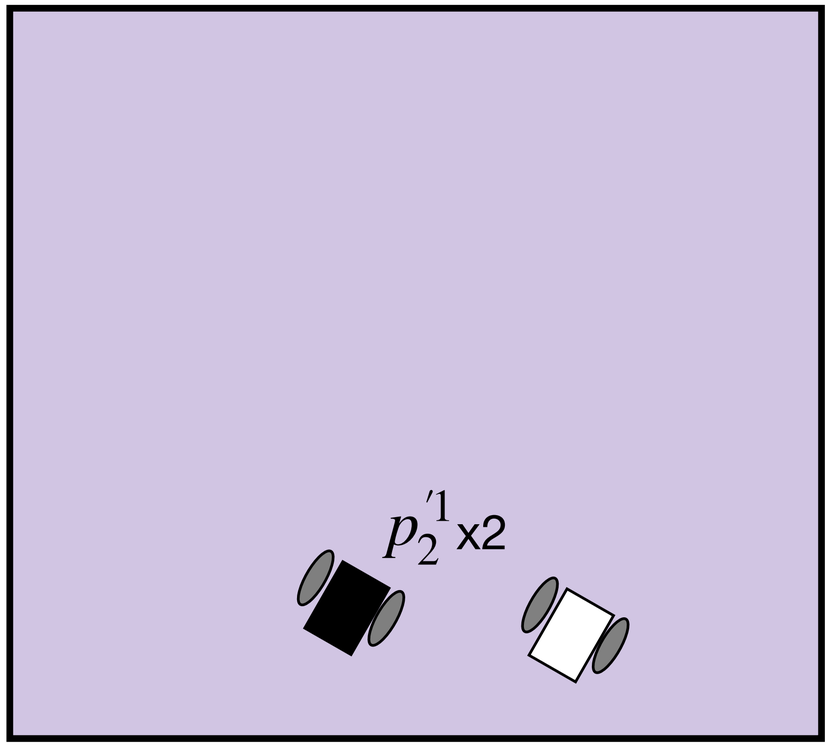
$$P_1 = \begin{bmatrix} p_1'^2 \\ \\ \end{bmatrix}$$



Algorithm Example

- **Time step t_1**
  - Correction: Resample from the temporary state distribution based on the weights $w_1^2 > w_1^1 > w_1^3$

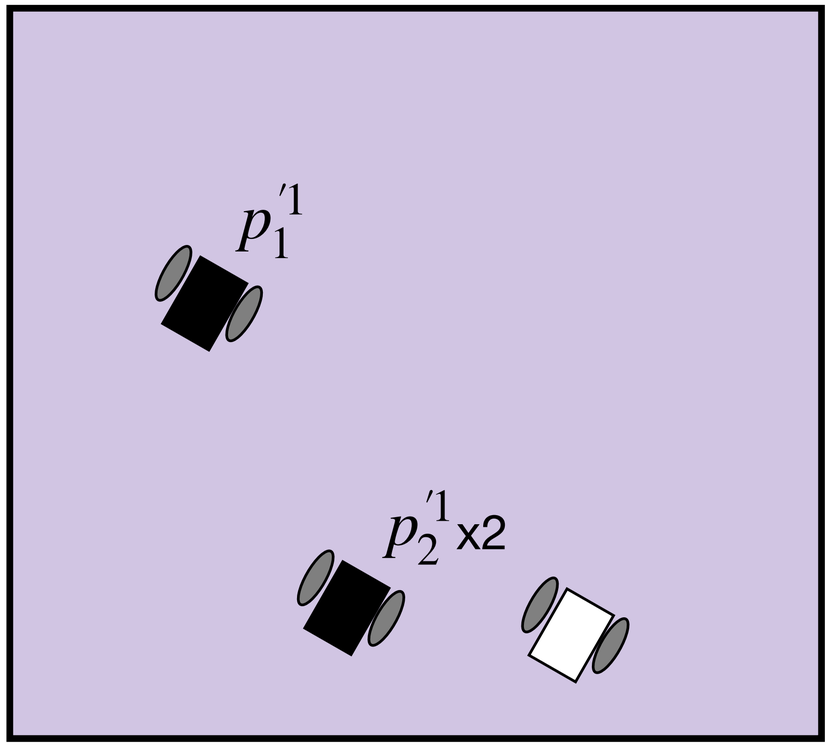
$$P_1 = \begin{bmatrix} p_1'^2 \\ p_1'^2 \\ \\ \end{bmatrix}$$



Algorithm Example

- **Time step t_1**
  - Correction: Resample from the temporary state distribution based on the weights $w_1^2 > w_1^1 > w_1^3$

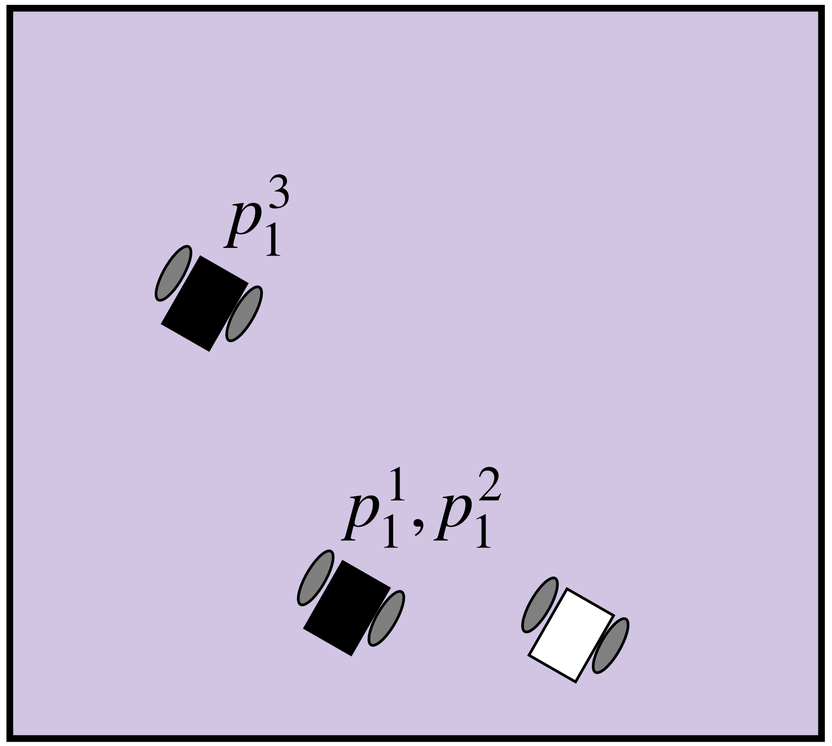
$$P_1 = \begin{bmatrix} p_1'^2 \\ p_1'^2 \\ p_1'^1 \end{bmatrix}$$



Algorithm Example

- **Time step t_1**
  - Correction: Resample from the temporary state distribution based on the weights $w_1^2 > w_1^1 > w_1^3$

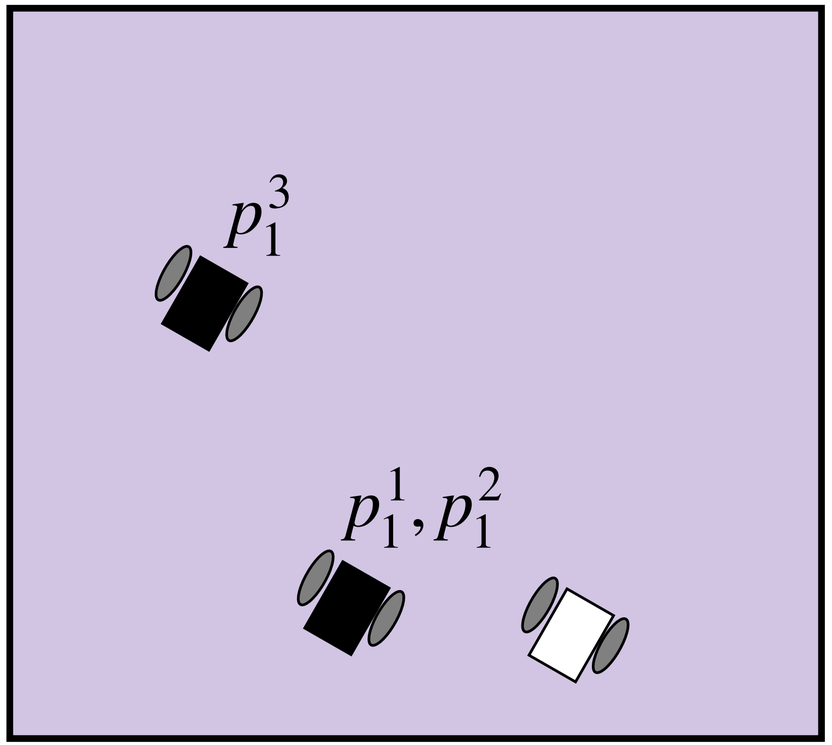
$$P_1 = \begin{bmatrix} p_1'^2 \\ p_1'^2 \\ p_1'^1 \end{bmatrix} = \begin{bmatrix} p_1^1 \\ p_1^2 \\ p_1^3 \end{bmatrix}$$



Algorithm Example

- **Time step t_1**
  - Correction: Resample from the temporary state distribution based on the weights $w_1^2 > w_1^1 > w_1^3$

$$P_1 = \begin{bmatrix} p_1^1 \\ p_1^2 \\ p_1^3 \end{bmatrix}$$



- Particle Filter Algorithm

Algorithm Particle_Filter(P_{t-1}, u_t, z_t)***:***

for $i = 1 \;..\; n$

x_t^i = sample from $P(x_t \mid x_{t-1}, u_t)$

$w_t^i = P(z_t \mid x_t)$

$p_t'^i = [x_t^i, w_t^i]^T$

$P_t' = [p_t'^1, p_t'^2, \ldots, p_t'^n]$

for $i = 1 \;..\; n$

p_t^i = sample from P_t' given weights

$P_t = [p_t^1, p_t^2, \ldots, p_t^n]$

return P_t

- Particle Filter Algorithm

Algorithm Particle_Filter(P_{t-1}, u_t, z_t)***:***

for $i = 1 \;..\; n$

x_t^i = sample from $P(x_t \mid x_{t-1}, u_t)$

$w_t^i = P(z_t \mid x_t)$

$p_t'^i = [x_t^i, w_t^i]^T$ Prediction Step

$P_t' = [p_t'^1, p_t'^2, \ldots, p_t'^n]$

for $i = 1 \;..\; n$ Correction Step

p_t^i = sample from P_t' given weights

$P_t = [p_t^1, p_t^2, \ldots, p_t^n]$

return P_t

Agenda

1. Particle Representation
2. Particle Filter Algorithm
3. **Resampling**
4. PF Examples

- Resampling Algorithm #2

Algorithm Resampling_2(P'_t):

$P_t^* = []$

for $i = 1 \; .. \; n$

$q = round(w^i / \Delta)$

for $j = 1 \; .. \; q$

$P_t^* = P_t^* \cup p_t^i$

$P_t = []$

for $i = 1 \; .. \; n$

$k = uniform \; sample \; from \; [1, \; len(P_t^*)]$

$P_t = P_t \cup p_t^{k*}$

return P_t

- **Resampling**
  - NOTE 1: Only resample with new measurements!

Agenda

1. Particle Representation
2. Particle Filter Algorithm
3. Resampling
4. **PF Examples**

